Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

R26-CS-010

Content

01 The Problem

02 The Proposed Solution

03 Individual Components

04 Ability of Commercialization

05 Work Breakdown Structure

06 Gantt Chart

THE PROBLEM



THE PROBLEM

Users and Stakeholders

- AV developers and automotive manufacturers who need large labelled datasets to train perception models.
- Pedestrians and road users whose faces and movement patterns are captured during road testing.
- Data protection regulators (GDPR, Sri Lanka PDPA) who must verify that AI training pipelines are lawful.
- Academic researchers who cannot legally share real driving footage for collaborative study.

Impact on Users and Society

- Individuals: personal location patterns and identities captured on public roads can be exposed through AI training pipelines without their knowledge or consent.
- AV industry: GDPR fines across EU member states exceeded EUR 1.6 billion in 2022 alone, creating direct financial and legal risk for companies using real driving data.
- Society: without privacy-safe training data, the development of trustworthy autonomous transport is slowed by legal and ethical barriers.
- Research community: inability to share training data limits reproducibility and collaboration in AV safety research.

Existing Solutions and Their Limitations

- Face and plate blurring: post-processing anonymisation degrades data quality and does not protect against membership inference attacks at the model level.
- Synthetic simulators (CARLA, AirSim, NVIDIA DRIVE Sim): reduce real-data need but carry no formal privacy certificate and do not protect the calibration channel.
- Differentially private GANs (DP-GAN, DP-CGAN): validated only on single-modality benchmarks (MNIST, CIFAR-10); not extended to multi-modal AV sensor data.
- No existing tool enforces privacy at the generation stage itself or produces an auditable privacy guarantee for synthetic AV datasets.

THE PROPOSED SOLUTION

Solution Overview

- A privacy-by-design pipeline that generates synthetic autonomous vehicle training data with formal epsilon-differential privacy guarantees.
- Real driving data is used only for calibration statistics. Those statistics are privatised before reaching the simulator. The generative model is trained with per-sample gradient clipping and Gaussian noise. No real individuals appear in the final synthetic dataset.

How It Overcomes the Limitations

- Existing tools use post-processing blurring: this fails against membership inference attacks at the model level. This solution enforces privacy at the generation stage itself.
- No existing simulator provides a formal privacy certificate. This pipeline outputs an epsilon-DP audit report validated by shadow-model MIA tests.
- Current DP-GAN research covers only single-modality benchmarks. This work extends epsilon-DP to multi-modal AV sensor data (camera, LiDAR, radar) and the calibration channel.

Individual Components and Contributions

IT22309556 | Privacy Integration

- Applies Laplace noise to Waymo calibration statistics
- Trains generative model under epsilon-DP via Opacus
- Tracks budget with RDP accountant; halts at epsilon = 10
- Produces Privacy Audit Interface report

IT22541284 | Dataset Generation

- Configures CARLA with privatised scene statistics
- Generates synthetic multi-sensor AV dataset batches
- Validates sim-to-real gap using FID score
- Delivers dataset to IT22110220

IT22110220 | Utility Evaluation

- Trains YOLOv8 / Faster R-CNN on synthetic data
- Measures mAP and FID against real-data baseline
- Produces privacy-utility trade-off analysis
- Delivers evaluation results to group thesis

IT22066916 | Risk and Compliance

- Maps project to GDPR Articles 5, 17, and 25
- Runs MIA and re-identification attack tests
- Verifies PDPA No. 9 of 2022 compliance
- Produces compliance and risk report



IT22309556 | S. A. S. D. PRIYADARSHI

Individual Component: **Privacy Integration**

Contribution:

- Builds the privacy enforcement layer for the group pipeline.
- Applies Laplace noise to real-world calibration statistics before simulator use and enforces per-sample gradient clipping with Gaussian noise via Opacus during generative model training.
- Tracks total privacy budget using the Renyi DP accountant and halts training at $\varepsilon = 10$.

Novelty:

- Privacy enforced during data generation rather than post-processing.

PRIVACY INTEGRATION

Privacy enforced during data generation rather than post-processing.

Problem:

- AV training data captures faces, license plates and location patterns, all qualifying as personal data under privacy acts like GDPR.
- Simulators must be calibrated against real driving records to produce useful data, reintroducing privacy risk.
- Membership inference attacks can extract individual records from a model calibrated on real data.

Main objective:

- To design an ε-differential privacy enforcement pipeline achieving a cumulative budget of epsilon <= 10, MIA attack success below 55%, and mAP within 15% of the real-data baseline.

sub-objectives:

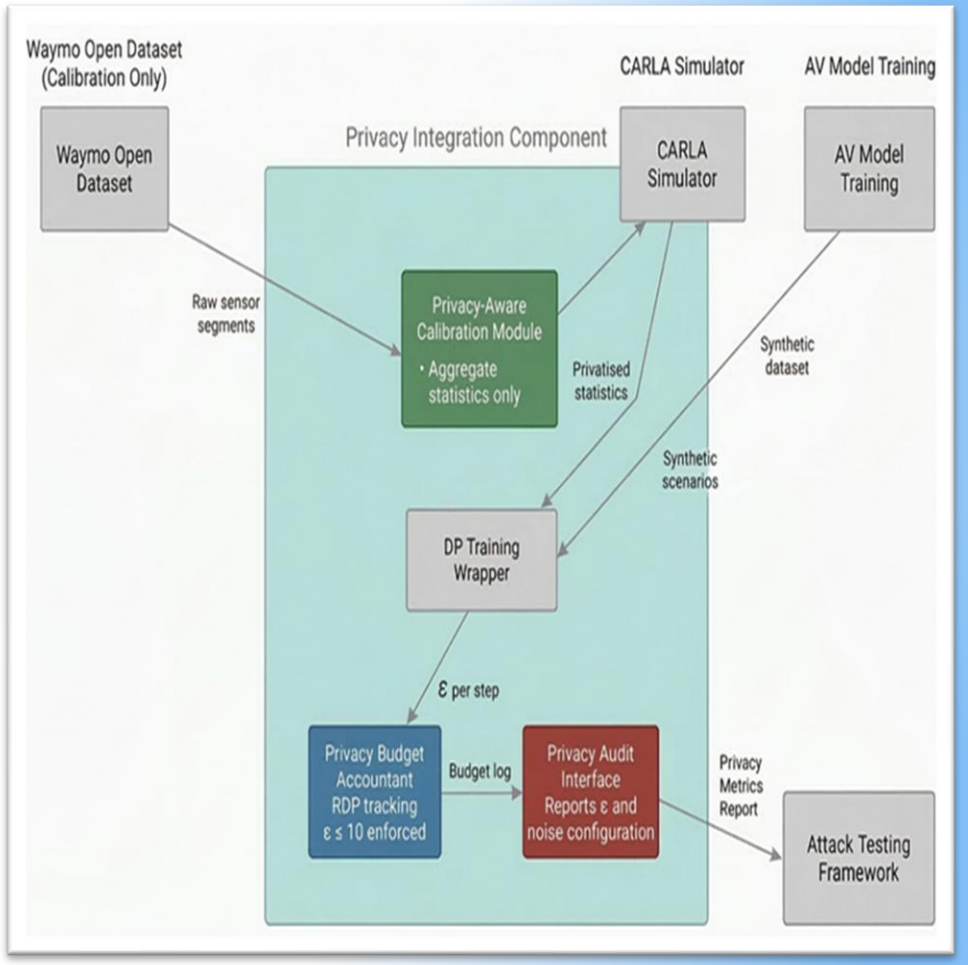
- To evaluate Opacus and TF Privacy against per-sample gradient support and RDP accounting accuracy and select one.
- To implement per-sample gradient clipping and Gaussian noise injection using Opacus, tested across at least three clipping norm and noise multiplier configurations.
- To apply the Laplace mechanism to all aggregate calibration statistics before they reach the simulator, keeping calibration epsilon below 20% of the total budget.
- To produce a quantitative privacy-utility trade-off analysis across at least three epsilon configurations, reporting epsilon, MIA rate, and synthetic data quality for each.

PRIVACY INTEGRATION

Application of Domain Knowledge

Research Methodology:

This component is based on knowledge from Differential Privacy Theory, Deep Learning Security, and Adversarial Machine Learning. These research areas study how to formally bound information leakage during model training and how to validate privacy guarantees through empirical attack-based testing.

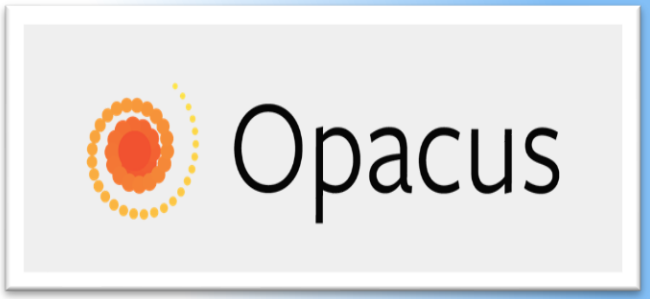


PRIVACY INTEGRATION

Research Methodology

Technologies Use:

- Opacus (PyTorch): per-sample gradient clipping and Gaussian noise injection.
- Renyi DP accountant: tracks cumulative epsilon at every training step.
- Laplace mechanism (NumPy): applied to aggregate calibration statistics.
- CARLA Simulator: generates synthetic multi-sensor AV training data.



Validation and Evaluation:

- Shadow-model MIA attack: target success rate at or below 55%.
- Re-identification test: synthetic trajectory match rate below 10%.
- AV model mAP: within 15% of real-data Waymo baseline.
- FID score: statistical similarity between synthetic and real sensor frames.



Data Availability, Security and Ethical Clearance:

- Waymo Open Dataset used under non-commercial research license.
- GDPR Article 25 (Privacy by Design) and Sri Lanka PDPA No. 9 of 2022 observed throughout.
- No real individuals appear in the CARLA-generated synthetic dataset.

PRIVACY INTEGRATION

Implementation Details

System Overview of the Individual Component:

- **Privacy-Aware Calibration Module:** reads Waymo aggregate statistics, adds Laplace noise, passes privatized values to CARLA. Raw sensor frames are never written to disk.
- **DP Training Wrapper:** clips per-sample gradients to L2 norm C, averages across the batch, then injects Gaussian noise before each weight update via Opacus.
- **Privacy Budget Accountant:** accumulates epsilon cost at every step using the RDP accountant. Training halts automatically when the ceiling is reached.
- **Privacy Audit Interface:** packages final epsilon, clipping norm, noise multiplier and budget log into a structured report delivered to IT22066916.

Dependencies and common component:

- IT22541284: receives privatized calibration statistics at end of Phase 1 to configure CARLA.
- IT22110220: receives the final synthetic dataset for AV model training and utility evaluation.
- IT22066916: receives the Privacy Audit Interface report to run MIA and re-identification tests.
- Shared tools: Opacus 1.x, PyTorch 2.x, NumPy. Waymo Open Dataset (non-commercial research license).

IT22541284 | YUBITHA S.



Individual Component: **Dataset Generation**

Contribution:

- Configures the CARLA autonomous driving simulator using privacy-preserved calibration statistics.
- Generates large-scale synthetic autonomous vehicle datasets containing multi-sensor outputs.
- Produces labeled datasets including RGB images, depth maps and semantic segmentation data.
- Ensures dataset diversity by simulating different driving environments, traffic density and weather conditions.
- Delivers the generated dataset batches to the utility evaluation component for model training.

Novelty

Generation of **privacy-aware synthetic multi-sensor datasets directly from the simulator environment**, enabling realistic AV training data while preserving privacy constraints.

DATASET GENERATION

Problem Definition

Problem:

Autonomous vehicle perception models require **large volumes of labeled driving datasets** for training.
However, collecting real-world driving data introduces **privacy concerns**, including exposure of faces, license plates and location information.
Additionally, acquiring and annotating real driving datasets is **expensive, time-consuming and legally restricted**.

Main objective:

To generate **realistic privacy-aware synthetic autonomous vehicle datasets** using the CARLA simulation platform for training machine learning perception models.

sub-objectives:

- Configure the CARLA simulator with appropriate sensors such as cameras and depth sensors.
- Generate diverse driving scenarios including different traffic densities, weather conditions and road environments.
- Produce labeled synthetic datasets including RGB images, depth maps and segmentation labels.
- Validate dataset realism and diversity to support machine learning model training.

Research Methodology

The research follows a **simulation-based dataset generation approach**.

First, the CARLA simulator is configured to create realistic driving environments.

Different driving scenarios are generated including variations in traffic, weather and road layouts.

Sensors such as **RGB cameras, depth sensors and segmentation cameras** capture simulation outputs. The generated sensor outputs are converted into **structured labeled datasets** suitable for autonomous vehicle perception model training.

Finally, the generated datasets are validated and delivered to the evaluation module for machine learning model training and performance analysis.

Technologies Used

1) CARLA Simulator – Synthetic autonomous driving environment generation
2) Python – Simulation control and data processing
3) OpenCV – Image processing and dataset handling
4) NumPy – Data manipulation
5) PyTorch – Integration with machine learning training pipeline

IMPLEMENTATION DETAILS

System Overview

- The dataset generation component operates within the CARLA simulation environment.
- The system configures the simulator using privacy-preserved calibration parameters received from the privacy integration module.
- It generates multiple autonomous driving scenarios and captures sensor outputs from the simulator.
- These outputs are converted into labeled datasets containing RGB images, depth maps and segmentation masks.
- The generated dataset batches are then delivered to the **utility evaluation module** for training perception models.

Dependencies and Common Components

- Privacy Integration Module (IT22309556) – provides privatized calibration statistics.
- Utility Evaluation Module (IT22110220) – receives generated datasets for training AV models.
- Risk and Compliance Module (IT22066916) – evaluates privacy risks of the generated system.

Shared technologies include **CARLA Simulator, Python, PyTorch and NumPy**.

IT22110220 | APEKSHA M.K.S.R



Individual Component: Utility Evaluation and Privacy–Utility Trade–off Analysis for Autonomous Vehicle Perception Models.

Contributions :

- Defines the evaluation framework to measure the performance of AV perception models that are trained using synthetic datasets.
- Defines the model training pipeline using tools such as PyTorch and YOLO.
- Utilizes synthetic datasets generated using simulation and dataset generation modules.
- Allows the model to recognize objects such as vehicles, pedestrians and traffic signs.
- Provide experimental results to assess the efficacy of privacy - enhanced synthetic datasets.

Novelty: Framework generating privacy–enhanced synthetic datasets evaluating AV perception privacy–utility trade–off.

UTILITY EVALUATION AND PRIVACY – UTILITY TRADE –OFF ANALYSIS FOR AUTONOMOUS VEHICLE PERCEPTION MODELS.

Framework generating privacy–enhanced synthetic datasets evaluating AV perception privacy–utility trade–off.

Problem Definition:

- The lack of evaluation methods to measure how privacy – enhanced synthetic datasets affect autonomous vehicle perception model performance.

Research Objective:

- To evaluate the effectiveness of privacy – enhanced synthetic datasets for training autonomous vehicle perception models by measuring the privacy utility trade – off using quantitative metrics.

Sub objectives:

- To create synthetic autonomous driving datasets with the aid of simulation platforms.
- To ensure maintenance of privacy while developing the synthetic datasets.
- To train the autonomous vehicle perception model with the synthetic dataset created in the previous step.

UTILITY EVALUATION AND PRIVACY – UTILITY TRADE –OFF ANALYSIS FOR AUTONOMOUS VEHICLE PERCEPTION MODELS.

Application of Domain Knowledge

Research Methodology:

- The research proposes an experimental approach to evaluate the effectiveness of privacy enhanced synthetic data in training models for autonomous vehicle perception.
- The synthetic data for driving scenarios is created through a simulator environment, and preserving techniques are utilized to develop privacy enhanced synthetic data.
- The data is utilized to train models to detect various objects such as cars, pedestrians and signs.
- Finally, the results are analyzed to assess the privacy utility trade off and the effectiveness of proposed approach.

Latest Technologies:

- CARLA Simulator: Synthetic dataset generation
- PyTorch: Deep learning model training
- YOLO: Object detection model
- Python: Data Processing and model development
- KITTI Dataset: Baseline evaluation dataset

UTILITY EVALUATION AND PRIVACY – UTILITY TRADE –OFF ANALYSIS FOR AUTONOMOUS VEHICLE PERCEPTION MODELS.

Application of Domain Knowledge

Validation and Evaluation:

- Test how well privacy enhanced synthetic datasets train autonomous vehicle models.
- Measure performance using Precision, Recall, F1 Score and mAP.
- Check the balance between privacy and model accuracy.
- Compare with models trained on normal synthetic datasets.

Data Availability, Security and Ethical Clearance:

- Synthetic datasets will be generated in simulation, ensuring availability and control.
- Data and models will be securely stored with restricted access.
- No real personal data will be used; research follows ethical guidelines for privacy.

UTILITY EVALUATION AND PRIVACY – UTILITY TRADE –OFF ANALYSIS FOR AUTONOMOUS VEHICLE PERCEPTION MODELS.

Implementation Details

System Overview:

- The system generates **privacy-enhanced synthetic datasets** for autonomous vehicle perception.
- It trains perception models to detect vehicles, pedestrians and traffic signs.
- The system evaluates model performance using metrics like Precision, Recall, mAP.
- It ensures a balance between data privacy and model accuracy.

Dependencies and Common Components:

- **Simulation Platform:** CARLA for generating synthetic driving datasets.
- **Deep Learning Framework:** Pytorch for training perception models.
- **Programming environment:** Python with libraties like Numpy for data handling.
- **Version control :** Git/GitHub for managing code and datasets.

IT22066916 | WIJESINGHE R.D.P.T.



Individual Component: Privacy Risk Evaluation using Membership Inference Attacks in Synthetic Autonomous Vehicle Data

Contribution to Overall Solution

- This component evaluates whether trained machine learning models leak information about the data used during training.
- It tests the privacy strength of models developed using synthetic autonomous vehicle datasets models using **Membership Inference Attacks (MIA)**.
- Verifies compliance with **GDPR privacy principles.**

Novelty

- Introduces an **attack-based privacy evaluation method** for synthetic AV data pipelines.
- Uses **Membership Inference Attacks (MIA)** to detect possible training data leakage.
- Provides **quantifiable privacy risk indicators**, enabling informed mitigation.

PRIVACY RISK EVALUATION USING MEMBERSHIP INFERENCE ATTACKS

Problem Definition

- Machine learning models can unintentionally memorize training data
- Attackers may identify whether a specific record was used during training
- This creates privacy risks when using sensitive datasets

Research Objective

- Develop a **privacy evaluation framework using Membership Inference Attacks** to detect potential leakage in trained models.

Sub-Objectives

- Implement a membership inference attack model
- Test trained ML models for privacy leakage
- Measure attack success using evaluation metrics
- Analyze privacy protection effectiveness
- Map results to GDPR privacy principles

PRIVACY RISK EVALUATION USING MEMBERSHIP INFERENCE ATTACKS

Research Methodology – Knowledge Domain

This component is based on knowledge from Machine Learning Security and Privacy-preserving Artificial intelligence. These research areas study how machine learning systems may expose sensitive information and how to evaluate privacy risks.

Technologies Used

- Python for system implementation
- Pytorch for machine learning model handling
- Scikit-learn for attack model development
- Attack simulation techniques for Membership Inference

These technologies allow the system to simulate attack scenarios and analyze the privacy behavior of trained models.

Purpose

- Simulate attack scenarios to analyze privacy behavior.
- Identify vulnerabilities in trained AV models without using sensitive personal data.

PRIVACY RISK EVALUATION USING MEMBERSHIP INFERENCE ATTACKS

Validation and Evaluation

The proposed system will be validated by running membership inference attacks against machine learning models.

Evaluation will be performed using metrics such as:

- Attack accuracy
- Privacy leakage risk
- Compliance indicators

The results will show whether the models are vulnerable to inference attacks and whether privacy-preserving mechanisms reduce these risks.

Data Availability, Security and Ethical Considerations

The research will use publicly available datasets and synthetic data generated from simulation environments. Sensitive personal data will not be directly used. Ethical guidelines for responsible data usage and privacy protection will be followed throughout the research process. Follow GDPR principles such as data minimization and privacy protection

PRIVACY RISK EVALUATION USING MEMBERSHIP INFERENCE ATTACKS

System Overview

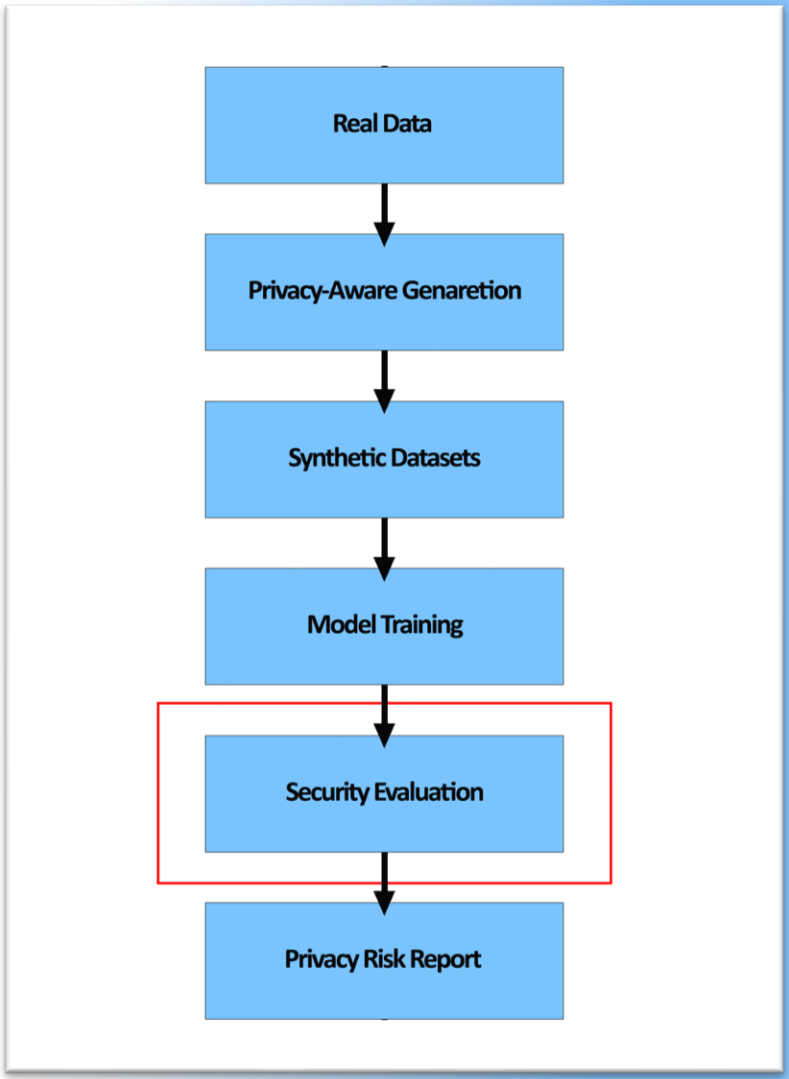
The component evaluates trained machine learning models by simulating membership inference attacks. The system attempts to determine whether specific data samples were used during model training.

Dependencies and Common Components

This module depends on outputs produced by other components in the research pipeline, **such as:**

- Synthetic dataset generation
- Privacy integration module
- Model training module

The evaluation results produced by this component help verify the overall privacy strength of the proposed system.



Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

Ability of Commercialization

Investment and Cost

- **Cloud GPU compute:** ~200 hours on AWS (AWS g4dn.xlarge - approx LKR 160/hr) (**LKR 32,000**).
- **Cloud storage:** 1 TB for dataset batches and model checkpoints (**LKR 8,000**).
- **Total cost: LKR 40,000.**

✓ No software licence fees (Opacus, PyTorch, CARLA are open source).

Recovery of Investment

- SaaS model: clients submit a calibration configuration and receive a certified synthetic dataset with an epsilon-DP audit report.
- Per-run pricing for occasional users; subscription tiers for teams generating data at scale, priced by compute volume.
- B2B license: embed the full pipeline into an existing simulator platform (CARLA or NVIDIA DRIVE Sim) under an annual agreement.

Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

Ability of Commercialization

Target Market

- AV manufacturers and automotive OEMs that need GDPR-compliant training data at scale.
- Academic research labs working on AV perception that cannot share real driving footage.
- Regulatory bodies and auditors that need to verify privacy guarantees on synthetic datasets.

Value Proposition

- Synthetic AV training data with a formal epsilon-DP certificate and MIA attack validation report.
- No real-world data licensing, consent infrastructure, or data retention liability.
- Faster regulatory approval: the audit report is ready to present to data protection authorities.

Customer Profile

- ML engineers and data infrastructure leads at AV companies under GDPR, CCPA, or PDPA.
- University research teams needing shareable, privacy-safe AV datasets for publication.
- Product safety and legal teams at automotive OEMs preparing regulatory submissions.

Intellectual Property Rights

- The pipeline combining RDP budget accounting with multi-modal AV simulator calibration under the Laplace mechanism is a candidate for a provisional patent.
- An Apache 2.0 open-source release is viable, with revenue drawn from the SaaS tier rather than the code itself.

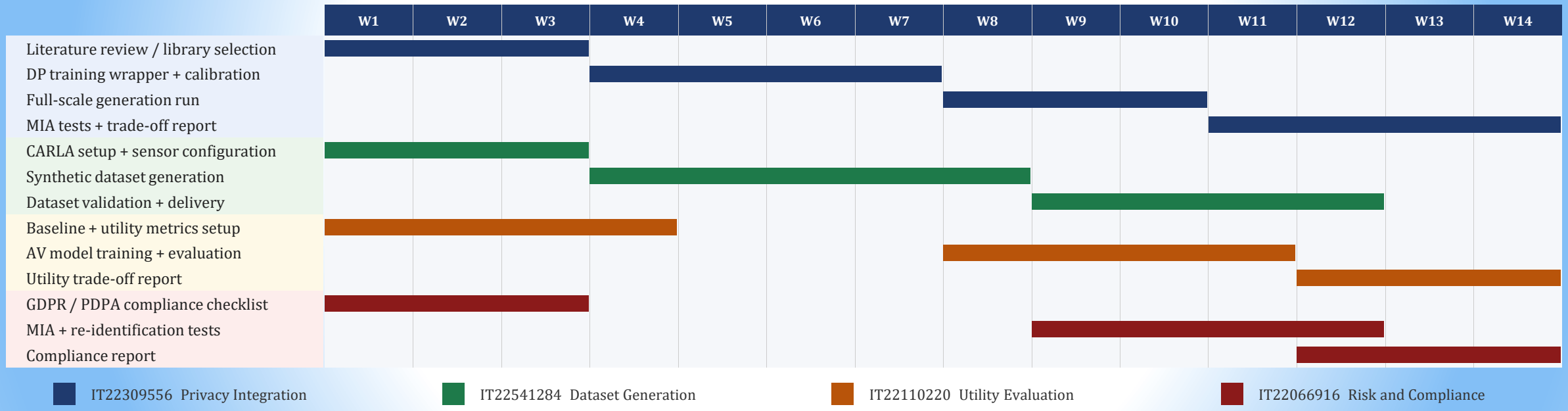
Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

Work Breakdown Structure

Member	Role	Weeks	Key Tasks
IT22309556	Privacy Integration	Wk 1-3	Literature review on DP mechanisms; library selection (Opacus vs TF Privacy)
IT22309556	Privacy Integration	Wk 4-7	Build DP training wrapper; implement Calibration Module; run ablation experiments
IT22309556	Privacy Integration	Wk 8-10	Full-scale generation run under epsilon-DP; monitor and log budget
IT22309556	Privacy Integration	Wk 11-14	Coordinate MIA tests with IT22066916; finalise trade-off report
IT22541284	Dataset Generation	Wk 1-3	Set up CARLA simulator; define scene and sensor configurations
IT22541284	Dataset Generation	Wk 4-8	Run CARLA using privatised calibration statistics; generate dataset batches
IT22541284	Dataset Generation	Wk 9-12	Validate dataset format and coverage; deliver final dataset to IT22110220
IT22110220	Utility Evaluation	Wk 1-4	Define utility metrics (mAP, FID); set up training pipeline and baseline
IT22110220	Utility Evaluation	Wk 8-11	Train AV model on synthetic dataset; measure mAP and FID vs baseline
IT22110220	Utility Evaluation	Wk 12-14	Report utility trade-off results; contribute to group thesis
IT22066916	Risk and Compliance	Wk 1-3	Review GDPR, CCPA, and PDPA requirements; define compliance checklist
IT22066916	Risk and Compliance	Wk 9-12	Run MIA and re-identification attack tests on DP synthetic data
IT22066916	Risk and Compliance	Wk 12-14	Compile attack results and compliance report; contribute to group thesis

Privacy-Enhanced Synthetic Data Generation for Autonomous Vehicle Training

Gantt Chart





Thank You!